Chapter Summary

Key Terms

6.1 Understanding Percent

- Percent
- Fractional form
- Decimal form
- Total
- Base
- Percent of the total
- Part
- Amount

6.2 Discounts, Markups, and Sales Tax

- Discount
- Cost
- Markup
- Retail price

6.3 Simple Interest

- Interest
- Principal
- Annual percentage rate
- Simple interest
- Term
- Due
- Origination date
- Payoff amount
- Future value
- Partial payment
- Present value

6.4 Compound Interest

- Compound interest
- Effective annual yield

6.5 Making a Personal Budget

- Budget
- Necessary expenses
- Fixed expenses
- Variable expenses
- 50-30-20 budget philosophy

6.6 Methods of Savings

- Savings account
- 1099 form
- Certificate of deposit
- Money market account
- Return on investment
- Ordinary annuity

6.7 Investments

- Bonds
- Maturity date
- Stocks
- Dividend
- Mutual fund

- Prospectus
- Issue price
- Shares
- Stock table
- Individual retirement account
- Roth IRA
- 401(k)

6.8 The Basics of Loans

- Fixed interest rate
- Variable interest rate
- Installment loan
- Loan amortization
- Revolving credit
- Amortization table
- Cost of finance

6.9 Understanding Student Loans

- FAFSA
- College funding gap
- Subsidized loan
- Unsubsidized loan
- Parent loan for undergraduate students
- Private student loan
- School-channel loan
- Direct-to-consumer loan
- Standard repayment plan
- Federal consolidation
- Refinancing
- Private consolidation
- Graduated repayment plan
- Extended repayment plan
- Discretionary income
- Pay as you earn (PAYE) repayment plan
- Revised pay as you earn (REPAYE) repayment plan
- Income-based (IBR) repayment plan
- Income-contingent (ICR) repayment plan
- Delinquent
- Default
- Rehabilitation

6.10 Credit Cards

- Reward program
- Annual fee
- Credit limit
- Bank-issued credit card
- Store-issued credit card
- Travel and entertainment cards
- Charge cards
- Billing period
- Balance
- Minimum payment
- Average daily balance

6.11 Buying or Leasing a Car

- Title and registration fees
- Destination fee
- Documentation fee

- Dealer preparation fee
- Extended warranty
- Down payment
- Acquisition fee
- Security deposit
- Disposition fees
- Liability insurance
- Collision insurance
- Comprehensive insurance
- Uninsured or underinsured motorist insurance
- Medical payment insurance
- Personal injury insurance
- Gap insurance
- Rental reimbursement insurance

6.12 Renting and Homeownership

- Lease
- Landlord
- Mortgage
- Escrow account
- Assessed value

6.13 Income Tax

- Gross income
- Adjusted gross income
- Exemption
- Taxable income
- Deduction
- Tax credit
- Earned income credit
- American opportunity credit
- Lifetime learning credit
- Child tax credit
- Child and dependent care tax credit
- Premium tax credit

Key Concepts

6.1 Understanding Percent

- Know what a percent is as a fraction, a decimal, and as a part of the whole.
- Use the percent equation to find any of the three values that are related by the equation.
- Apply the percent equation in applications.

6.2 Discounts, Markups, and Sales Tax

- Discounts are markdowns from an original price.
- Mark-ups are increases to the price paid by a retailer to cover their costs.
- be able to calculate the markup based on a percentage of the cost
- Sales taxes vary from state to state and often county to county.
- Retail prices, sales prices and percent discounts can be calculated if the other two values are known.
- Original costs, retail prices, and percent markup can be calculated if the other two values are known.
- In calculations, sales tax acts like a markup.

6.3 Simple Interest

- Interest is money that is paid by a borrower for the privilege of borrowing the money.
- Simple interest is computed by substituting the principal, interest rate, and number of years into the formula $I = P \times r \times t$
- The payoff for a loan is the amount of principal remaining on a loan plus the interest that accumulated on the loan since the last payment.

- The future value of an investment yielding simple interest is the original principal plus the interest earned on the investment.
- When making a partial payment, some of the payment pays off all the accumulated interest, while the remainder of the payment is applied to the principal of the loan.
- Finding the present value of an investment is used to determine how much should be invested now in order to achieve a specific goal.

6.4 Compound Interest

- Compound interest means that the interest earned during one period will earn interest in later periods. Essentially, the amount of the principal grows from period to period.
- The important values in computing compound interest are the interest rate, the principal, the length of time the investment, and the number of times the investment is compounded.
- Compound interest has minimal impact early, but later has a very large impact.
- You can determine how much to invest today in order to reach a goal for some time later.
- Compound interest can be translated into an effective annual yield, which allows for comparison between investment options.

6.5 Making a Personal Budget

- A budget is a set of guidelines for how to allocate your income.
- Budgeting helps to plan for many of life's expenses
- Budgets are used to compare income to expenses. When expenses exceed income, changes have to be made.
- Budgets can help evaluate the affordability of life changes.
- One guideline for setting a budget is the 50-30-20 budget philosophy. The guidelines suggest that 50% of income is allocated to necessary expenses, 30% to expenses that wants, and 20% to savings and other debt reduction.

6.6 Methods of Savings

- There are three main types of savings accounts, saving accounts, certificates of deposit (CD), and money market accounts.
- Savings account are very risk free, and so yield low interest rates.
- The differences in the three types of savings accounts relate to their convenience.
- Savings account typically have a lower interest rate that money market accounts, which typically have lower interest rates than CDs.
- Ordinary annuities more accurately reflect how we save, in that money is deposited repeatedly over time.
- Spreadsheet software, such as Google Sheets, have built in functions that can be used to quickly calculate both the future value of an ordinary annuity account, but also the payment necessary to reach a goal using an ordinary annuity.

6.7 Investments

- There are many different investments with different returns and risks.
- Bonds are loans form the purchaser to the entity selling the bond.
- Bonds have some tax benefits, low to no risk, and a low return.
- Stocks represent part ownership in a company. As such, stock holders share in the profits, and losses, of the company.
- Information, including price, P/E, yearly highs and lows, and dividend amount can be found in online stock tables available on many websites.
- Mutual funds represent collections of professionally administered investment vehicles. Have shares in a mutual fund has lower risk than ownership of stocks.
- Retirement accounts employ some of the same strategies as mutual funds, in that they spread the risk and are professionally managed.
- IRAs and Roth IRAs differ on when taxes are paid on the money, and who can use them. Roth IRAs have income limits while traditional IRAs do not.

6.8 The Basics of Loans

- There are many reasons for a loan, but primarily it is taken out for a large expense when cash is not available.
- Each payment for an installment loan consists of an interest portion and a principal portion.
- There is a formula to calculate the payment necessary to pay off a loan in installments.
- Amortization schedules, or tables, show how each payment is applied to principal and interest. It also includes other details such as remaining balance and total interest paid.

- Loans often have other fees associated with them such as origination fees or application fees. The total of the interest paid and the fees is the cost of finance.

6.9 Understanding Student Loans

- The FAFSA must be filled out each year that a student wishes to borrow for.
- A student's funding gap determines how much they need in loans to pay for college.
- Federal subsidized student loans defer payments until after graduation and interest does not accrue on these loans.
- Unsubsidized student loans defer payment until after graduation but interest begins accruing as soon as the loan finds are disbursed.
- There are both yearly and aggregate limits for student loans to prevent over-borrowing, among other reasons.
- Federal direct loans have a low interest rate set by the government, but other student loans have varying rates of interest set by the banks.
- The standard repayment plan lasts 10 years and is made up of monthly payments.
- Consolidating or refinancing student loans merges many student loans into one loan.
- If only federal loans are consolidated, the interest rate is the same as the individual loans, currently set at 4.99%.
- If other loans are refinanced together, the interest rate may be lower with the new loan.
- Other repayment plans are available. Such a plan may have payment that start small and grow as the loan is paid off, or it may have a longer term, or may be based on the discretionary income of the student.
- Being delinquent on a student loan is a precursor to being in default. Making payments in a timely fashion allows the student to avoid this situation.

6.10 Credit Cards

- Credit cards can be a flexible way to pay for almost anything, but can become a financial hazard if used unwisely.
- When deciding which credit card to apply for, evaluate the interest rate, fees (annual and late), reward programs and credit limit. Be sure they meet your criteria.
- Paying off the balance of your credit card every month will control your spending and will never result in paying interest.
- Credit card statements hold all important information about your credit card, including payment, balances, charges and billing cycle dates.
- Although the minimum payment is attractive precisely because it is so small, paying only the minimum results is a long payoff term and higher interest costs.

6.11 Buying or Leasing a Car

- There are many factors to consider when choosing to buy or lease a car.
- The cost of the car is increased by a number of fees and sales tax.
- There are advantages to buying a car and advantages to leasing a car. The decision between the two depends on the preference of the buyer.
- Insurance covers costs associate with accidents. It is made up of various components.
- The costs of owning a car, including insurance and maintenance, should be a part of the budgeting process.
- Budgeting for unexpected repairs can ease the stress of encountering large repair bill.

6.12 Renting and Homeownership

- There are many points of comparison between renting and buying a house.
- Before deciding to buy a house, you should carefully consider all the responsibilities that come with home ownership.
- Renting comes with more restrictions on the renter, but with fewer costs and is easier to move from.
- Owning a house has more costs but has more freedom, plus the owner creates equity.
- Mortgages are loans, and payments are calculated in the same way as any other loan.
- Amortization tables help a homeowner understand the mortgage and how the payments are applied to the principal and interest.
- In addition to paying the amount financed for a mortgage, the monthly payment will include an escrow payment, which covers insurance and taxes.

6.13 Income Tax

- Federal income tax is based on income after certain adjustments.
- Gross income is income from all sources, including gifts and winnings.
- Before taxes are calculated, the taxable income is found by subtracting deductions and exemptions from gross income.

- Income tax is progressive, increasing in rate as income increases.
- Being in the 32% tax bracket means some of your income is taxed at 10%, some at 12%, some at 22%, some at 24%, and the rest at 32%.
- Income in each tax bracket is taxed at that bracket's rate, which means in 2022 the first $10,275 earned is taxed at 10% only.
- Tax credits are subtracted from the taxes that are owed.
- Some tax credits are refundable, which means they can make the amount you owe negative, which results in a refund.

Videos

6.1 Understanding Percent

- Finding Percent of a Total (https://openstax.org/r/solve_percent_problem1)
- Finding the Total from the Percent and the Part (https://openstax.org/r/solve_percent_problem2)
- Finding the Percent When the Total and the Part Are Known (https://openstax.org/r/solve_percent_problem3)

6.2 Discounts, Markups, and Sales Tax

- Computing Price Based on a Percent Off Coupon (https://openstax.org/r/Computing_Price_Based)
- Finding Sales Tax Percentage (https://openstax.org/r/Finding_Sales_Tax)

6.4 Compound Interest

- Compound Interest (https://openstax.org/r/compound_interest_beginners)
- Compare Simple Interest to Interest Compounded Annually (https://openstax.org/r/compare_simple_compound_interest1)
- Compare Simple Interest and Compound Interest for Different Number of Periods Per Year (https://openstax.org/r/compare_simple_compound_interest2)

6.5 Making a Personal Budget

- Creating a Budget (https://openstax.org/r/creating_budget)
- 50-30–20 Budget Philosophy (https://openstax.org/r/50-30-20_budgeting_rule)

6.6 Methods of Savings

- Return on Investment, ROI (https://openstax.org/r/This_video)
- Future Value Using Google Sheets (https://openstax.org/r/cell_references)

6.7 Investments

- Bonds (https://openstax.org/r/investing_basics_bonds)
- Reading Stock Summary Online (https://openstax.org/r/Reading_Stock)
- Mutual Funds (https://openstax.org/r/investing_basics_mutual_funds)
- 401(k) Accounts (https://openstax.org/r/401(k)s)

6.8 The Basics of Loans

- Credit Scores Explained (https://openstax.org/r/credit_scores_explained)
- Reading an Amortization Table (https://openstax.org/r/Amortization_Table)

6.9 Understanding Student Loans

- Types of Student Loans (https://openstax.org/r/Student_Loans)
- Repayment Plans (https://openstax.org/r/student_loan_repayment_plans)

6.10 Credit Cards

- Choosing a Credit Card (https://openstax.org/r/Credit_Card)
- Reading Credit Card Statements (https://openstax.org/r/Reading_Credit)

6.12 Renting and Homeownership

- Rent or Buy (https://openstax.org/r/Rent_or_Buy)
- Closing Costs (https://openstax.org/r/closing_costs)

6.13 Income Tax

- Deductions Versus Credits (https://openstax.org/r/Versus_Credits)
- Tax Brackets (https://openstax.org/r/Brackets)

Formula Review

6.1 Understanding Percent

part = percent x total

6.2 Discounts, Markups, and Sales Tax

$\text{discount} = \text{percent discount} \times \text{original price}$

$\text{sale price} = \text{original price} - \text{discount}$

$\text{sale price} = \text{original price} - \text{percent discount} \times \text{original price} = \text{original price} \times (1 - \text{percent discount})$

$\text{markup} = \text{percent markup} \times \text{cost}$

$\text{retail price} = \text{cost} + \text{markup}$

$\text{retail price} = \text{cost} + \text{percent markup} \times \text{cost} = \text{cost} \times (1 + \text{percent markup})$

$\text{sales tax} = \text{purchase price} \times \text{tax rate}$

$\text{Total price} = \text{purchase price} + \text{purchase price} \times \text{tax rate} = \text{purchase price} \times (1 + \text{tax rate})$

6.3 Simple Interest

$I = P \times r \times t$

$T = P + I$

$T = P + P \times r \times t$

$I = P \times \frac{r}{365} \times t$

$FV = P + I = P + P \times r \times t$

$A = P \times \frac{r \times (1 + r)^t}{(1 + r)^t - 1}$

$PV = \frac{FV}{(1 + rt)}$

6.4 Compound Interest

$A = P\left(1 + \frac{r}{n}\right)^{nt}$

$PV = \frac{A}{\left(1 + \frac{r}{n}\right)^{n \times t}}$

$Y = \left(1 + \frac{r}{n}\right)^n - 1$

6.6 Methods of Savings

$A = P\left(1 + \frac{r}{n}\right)^{nt}$

$\text{ROI} = \frac{FV - P}{P}$

$FV = pmt \times \frac{(1 + r/n)^{n \times t} - 1}{r/n}$

$pmt = \frac{FV \times (r/n)}{(1 + r/n)^{n \times t} - 1}$

6.7 Investments

$\text{annual return} = \left(\frac{FV}{P}\right)^{\left(\frac{1}{t}\right)} - 1$

$\text{P/E} = \frac{\text{Share Price}}{\text{Dividend}}$

$$\text{Yld\%} = \frac{\text{Annual Dividend}}{\text{Share Price}} \times 100\%$$

6.8 The Basics of Loans

$$I = P \times \frac{r}{n}$$

$$pmt = \frac{P \times (r/n) \times (1 + r/n)^{n \times t}}{(1 + r/n)^{n \times t} - 1}$$

6.9 Understanding Student Loans

$$\text{funding gap} = \text{total cost} - \text{all aid}$$

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

$$pmt = \frac{P \times (r/n) \times (1 + r/n)^{n \times t}}{(1 + r/n)^{n \times t} - 1}$$

$$\text{discretionary income} = \text{gross income} - 1.5 \times \text{poverty guideline}$$

6.10 Credit Cards

$$I = \frac{\text{ADB} \times r \times d}{365}$$

6.11 Buying or Leasing a Car

$$pmt = \frac{P \times (r/n) \times (1 + r/n)^{n \times t}}{(1 + r/n)^{n \times t} - 1}$$

$$\text{MD} = \frac{P - R}{n}$$

$$\text{APR} = 2400 \times \text{MF}$$

$$\text{MF} = \text{APR}/2{,}400$$

$$PMT = \frac{(P - R)}{n} + (P + R) \times \text{MF}$$

6.12 Renting and Homeownership

$$pmt = \frac{P \times (r/12) \times (1 + r/12)^{12 \times t}}{(1 + r/12)^{12 \times t} - 1}$$

$$T = pmt \times 12 \times t$$

$$\text{CoF} = T - P$$

Projects

Creating Your Future Budget

In this project, you will create a budget based on a job you are likely to have after you graduate.

1. Go online and research the average starting salary for the profession you are studying for. Use at least two sources. Be sure to record the web address from your search.
2. Approximate your monthly take-home pay. You may use the SmartAsset (https://openstax.org/r/paycheck_calculator) website to estimate this.
3. Use the 50-30-20 budget philosophy to determine how much you should budget for needs, wants and savings, or extra debt reduction.
4. Create a list of likely expenses. This list must include rent/mortgage, utilities, food, and school loan repayment. You may also want to include car payments, gasoline, and other items.
5. Categorize each expense as need, want, or savings.
6. Using the amounts found in step 3, decide how much to allocate to each of your expenses. It may help to quickly research how much rent is where you want to live.
7. Discuss the choices you had to make, and why you prioritized some expenses over others.

Interest Rate and Time: What Is the Relationship?

The interplay between interest rate and time for an annuity is not easily seen. How the amount that must be deposited per compounding period, pmt, changes based on the time and interest rate would be useful to understand. In this project, you will explore this relationship. We will use a fixed future value of FV = \$1,000,000 and a fixed number of periods per year, 12 (monthly compounding). With those, we'll find various annuity payments that must be made to reach the goal.

The annual interest rate for the investment is in the top row. The number of years for the investment is in the left column. In each cell (or box), find the monthly payment necessary to reach the goal of $1,000,000.

		Annual Interest Rate					
		1.5%	2.0%	3.0%	5.0%	7.5%	10.0%
Number of Years	10						
	15						
	20						
	30						
	40						
	45						

Describe how the interest rates and number of years impact the payment necessary to reach the goal of $1,000,000.

Finding a Home

In this project you will identify a home you like, and then estimate the costs associated with that home.

1. Find a home in your region that you would like to buy using an online search of listings in your area. Zillow is a good place to begin.
2. Find the asking price for this home. Assume you would pay that price.
3. Find an estimation for closing costs in your area. Assume you finance those costs also.
4. Estimate the taxes to be paid on the home per year. It is likely that the online listing of the home has an estimate for the taxes for the house.
5. Use Google to determine the average homeowner's insurance cost in your region.
6. Use the internet to determine the average interest rate for a 30-year mortgage.
7. Find how much you would pay per month, based on the answers to the previous questions, including the escrow payments for taxes and insurance.
8. Assume you will pay $50 per $100,000 borrowed in PMI. Add this to the monthly payment.

Chapter Review

Understanding Percent

1. Convert 0.45 to percent form.
2. What is 70% of 200?
3. Rewrite the fraction $\frac{240}{100}$ as a percentage.
4. 20 is what percent of 500?

Discounts, Markups, and Sales Tax

5. An item with a retail price of $250.00 is on sale for 30% off. What is the sale price of the item? Round to the nearest penny.
6. The sales price of an item is $38.50 after a 40% discount. What was the retail price of the item? Round to the nearest penny.
7. A retailer buys an item for $1,000.00 and has a 65% markup. What is the retail price of the item? Round to the nearest penny.
8. The sale price of an item is $46.00 and the retail price is $65.00. What is the percent discount? Round to two decimal places.
9. The retail price of an item is $345.38 and sales tax in the region is 8.25%. How much is the sales tax?

Simple Interest

10. $3,000 is invested in a 5-year CD earning 2.25% interest. How much is the CD worth in 5 years?
11. A $4,000.00 simple interest loan is taken out for 3 years at 12.5%. How much is owed when the loan comes due?
12. A $10,000 loan with annual simple interest of 14.9% is taken out for 90 days. How much is due in 90 days?
13. A simple interest loan for $20,000 is taken out at 13.9% annual interest rate. A partial payment of $13,500.00 is made 30 days into the loan period. After this payment, what will the remaining balance be?
14. Find the present value of an investment with future value of $15,000 if the investment earns 3.55% simple interest for 10 years?

Compound Interest

15. Find the future value after 20 years of $15,000 deposited in an account bearing 4.26% interest compounded quarterly.
16. What is the effective annual yield for an account bearing 3.21% interest compounded quarterly? Round to two decimal places if necessary.
17. Find the present value of $100,000 in an account bearing 5.25% interest compounded daily after 30 years.

Making a Personal Budget

18. In budgeting, what is the difference between an expense that is a need and an expense that is a want?
19. Apply the 50-30-20 budget philosophy to a monthly income of $6,000.00.
20. Create the budget for a person with the following income and bills, and determine how much the income exceeds or falls short of the expenses:
    Job = $5,250, Side job = $550, rent = $1,150.00, utilities = $150.00, internet = $39.99, student loans = $375.00, food = $550.00, bus and subway = $112.00, credit cards = $200.00, entertainment = $400.00, clothing = $150.00.
21. In question 20, the budget for a person was created. Can the person afford to buy a car if the monthly payments will be $365.50, monthly car insurance will be $114.75, and gasoline will be $200.00?

Methods of Savings

22. Why is a CD less flexible than a money market account?
23. Which is likely to have the highest interest rate, a savings account, CD, or money market account?
24. $3,000 is deposited in a money market account bearing 3.88% interest compounded monthly. How much is the account worth in 15 years?
25. Refer to question 24. What is the return on investment for the money market account? Round to two decimal

places, if necessary.

26. $300 is deposited monthly in an ordinary annuity that bears 6.5% interest compounded monthly. How much is in the annuity after 30 years?

27. How much must be deposited quarterly in an account bearing 4.89% interest compounded quarterly for 30 years so the account has a future value of $500,000?

Investments

28. Why is investing in a stock riskier than in a mutual fund?

29. Which investment offers high rates of return with reduced risk?

30. Which investment offer a fixed rate of return over time?

31. Which is the riskiest investment?

32. How much is earned on a 10-year bond with issue price of $1,000 that pays 3.9% annually?

33. Henri's employer matches up to 6% of salary for IRA contributions. Henri's annual income is $52,500. If Henri wants to deposit $3,000 annually in his IRA, how much in matching funds does the employer deposit in the IRA?

34. How much must be deposited quarterly in a mutual fund that is expected to earn 11.2% interest compounded quarterly if the account is to be worth $1,000,000 after 38 years?

35. Lorraine purchases stocks for $5,000. The stocks paid dividends by reinvesting the dividends in stock. She sold the stock for $13,750 after 6 years. What was Lorraine's annual return on that investment? Round to two decimal places if necessary.

36. What are two ways stocks earn money?

The Basics of Loans

37. What is an installment loan?

38. An installment loan with monthly payments has an outstanding balance of $3,560. If the annual interest rate is 9%, how much interest will be paid that month?

39. A loan for $42,000 is taken out for 6 years at 7.5% interest. What are the payments for that loan?

40. The total interest paid on a loan was $4,500. The loan had an origination fee of $400.00, a $125.00 processing fee, and a filing fee of $250.00. How much was the cost of financing for that loan?

41. Using the amortization table below, what is the remaining balance after payment 25?

Amortization Schedule Calculator					
Loan Amount		$17,500.00			
Interest Rate (annual)		8.25%			
Term (Years)		5			
Monthly Payment		$356.93			
Month	Payment	Principal	Interest	Total Interest	Balance
					$17,500.00
1	$356.93	$236.62	$120.31	$120.31	$17,263.38
2	$356.93	$238.25	$118.69	$239.00	$17,025.13
3	$356.93	$239.89	$117.05	$356.05	$16,785.24
4	$356.93	$241.54	$115.40	$471.44	$16,543.71
5	$356.93	$243.20	$113.74	$585.18	$16,300.51
6	$356.93	$244.87	$112.07	$697.25	$16,055.64
7	$356.93	$246.55	$110.38	$807.63	$15,809.09
8	$356.93	$248.25	$108.69	$916.32	$15,560.84
9	$356.93	$249.95	$106.98	$1,023.30	$15,310.89
10	$356.93	$251.67	$105.26	$1,128.56	$15,059.22
11	$356.93	$253.40	$103.53	$1,232.09	$14,805.82
12	$356.93	$255.14	$101.79	$1,333.88	$14,550.67
13	$356.93	$256.90	$100.04	$1,433.92	$14,293.77
14	$356.93	$258.66	$98.27	$1,532.19	$14,035.11
15	$356.93	$260.44	$96.49	$1,628.68	$13,774.66
16	$356.93	$262.23	$94.70	$1,723.38	$13,512.43
17	$356.93	$264.04	$92.90	$1,816.28	$13,248.39
18	$356.93	$265.85	$91.08	$1,907.36	$12,982.54
19	$356.93	$267.68	$89.25	$1,996.62	$12,714.86
20	$356.93	$269.52	$87.41	$2,084.03	$12,445.34
21	$356.93	$271.37	$85.56	$2,169.59	$12,173.97
22	$356.93	$273.24	$83.70	$2,253.29	$11,900.73
23	$356.93	$275.12	$81.82	$2,335.11	$11,625.62
24	$356.93	$277.01	$79.93	$2,415.03	$11,348.61
25	$356.93	$278.91	$78.02	$2,493.06	$11,069.69
26	$356.93	$280.83	$76.10	$2,569.16	$10,788.86

Understanding Student Loans

42. If a college program takes 5 years, how long is the student's eligibility period for student loans?

43. What is the funding gap for a student with cost of college equal to $35,750 and non-loan financial aid of $24,150?

44. What is the maximum amount of federal subsidized and unsubsidized loans for a dependent student in their third year of an undergraduate data science program?

45. A single person has gross income of $28,500. Their poverty guideline is $12,000. What is their discretionary income?

Credit Cards

46. The billing cycle for a credit card goes from March 15 to April 14. The balance at the start of the billing cycle is $450.00. The list of transactions on the card is below. Find the average daily balance for the billing cycle.

Date	Activity	Amount
15-Mar	Billing Date Balance	\$450.00
21-Mar	Movie	\$50.00
28-Mar	Gasoline	\$65.00
31-Mar	Snacks	\$15.50
1-Apr	Dinner	\$63.60
1-Apr	Payment	\$300.00
1-Apr	Gasoline	\$48.90
9-Apr	Plane Tickets	\$288.50

47. The average daily balance for Greg's last credit card statement was \$1,403.50. The card charges 15.9% interest. If the billing cycle for that statement was 30 days, how much interest is Greg charged?

48. The billing cycle for a credit card goes from March 15 to April 14. The balance at the start of the billing cycle is \$450.00. The list of transactions on the card is below. The interest rate is 15.9%. What is the balance due at the end of the billing cycle?

Date	Activity	Amount
15-Mar	Billing Date Balance	\$450.00
21-Mar	Movie	\$50.00
28-Mar	Gasoline	\$65.00
31-Mar	Snacks	\$15.50
1-Apr	Dinner	\$63.60
1-Apr	Payment	\$300.00
1-Apr	Gasoline	\$48.90
9-Apr	Plane Tickets	\$288.50

Buying or Leasing a Car

49. Name three common fees when leasing a car.

50. Name three common fees when buying a car.

51. If the cost of a car is \$35,500, and the residual value of the car after 3 years is \$27,800, what is the monthly depreciation for the car?

52. What is the annual interest rate if the money factor is 0.000045?

53. What is the payment for a lease on a car if the price of the car is \$29,900, the residual price of the car is \$16,500, the lease is for 24 months, and the APR is 8.9%?

Renting and Homeownership

54. Name two advantages of renting a residence.
55. Name two advantages of buying a home.
56. What is PMI?
57. What are the monthly payments for a 25-year mortgage for $165,000 at 4.8% interest, not including escrow?
58. What are the monthly payments, including escrow, for a 25-year mortgage for $165,000 at 4.8% interest, if the property taxes are $5,200 annually, the homeowner's insurance is $825 every 6 months, and the PMI is $45 per month?

Income Tax

59. What is the maximum gift that a person can receive that is an exemption on federal income taxes?
60. Find the gross income if a person earns $51,500 at their full-time job, they earn $5,300 for contracted services, and $5,000 in gambling winnings.
61. Find adjusted gross income for a person with a gross income of $67,850, if they had a $2,000 gift, they paid $1,000 in mortgage interest, $4,250 in property taxes, and they contributed $750 to a qualifying charity.
62. How much SSI is paid by a person with an annual income of $183,500?
63. How much in FICA taxes are paid from a person's paycheck with gross income of $3,450?
64. How much in federal income tax does a person filing single pay if their taxable income is $205,240? Use the tax table below.

Bracket	Lower Income Limit	Upper Income Limit	Tax Rate
1	0	$10,275	10%
2	$10,276	$41,775	12%
3	$41,776	$89,075	22%
4	$89,076	$170,050	24%
5	$170,051	$215,950	32%
6	$215,951	$539,900	35%
7	$539,901		37%

Chapter Test

1. 45 is what percent of 180?
2. Find 47% of 31. Round to two decimal places.
3. What is the sale price of a $90 shirt if there is a 25% discount for the shirt?
4. A small boutique buys purses for $27.50 per purse. What is the boutique's retail price for the purses if they add a 90% markup?
5. What was the discount on a computer is the retail price was $1,200 and the sale price is $950? Round to two decimal places if necessary.
6. A 15% simple interest loan of $2,000 is taken out for 75 days. How much is owed when the loan is due?
7. What is the future value of a $15,000 investment made at 4.65% interest compounded yearly for 10 years?
8. Apply the 50-30-20 budget philosophy to a monthly income of $3,920.

9. $2,500 is deposited in an account that bears 5.25% interest compounded quarterly for 15 years. What is the return on investment for that investment? Round to two decimal places if necessary.

10. How much per month must be paid into an ordinary annuity so its future value is $150,000 after 10 years if the annuity earns 3.85% interest compounded monthly?

11. Bethanie's employer matches up to 7% of annual salary for deposits into the company sponsored IRA. Bethanie earns $62,400 annually. She deposits $8,000 annually inter her IRA. How much does the company deposit in Bethanie's IRA?

12. If $360 is deposited monthly in a mutual fund that is expected to earn 9.35% compounded monthly, how much will the mutual fund be worth after 23 years?

13. A $14,250, 4-year loan is taken out at 9.15% interest. What are the monthly payments for the loan?

14. Using the amortization table below, how much total interest has been paid after payment 20?

Amortization Schedule Calculator					
Loan Amount		$17,500.00			
Interest Rate (annual)		8.25%			
Term (Years)		5			
Monthly Payment		$356.93			
Month	Payment	Principal	Interest	Total Interest	Balance
					$17,500.00
1	$356.93	$236.62	$120.31	$120.31	$17,263.38
2	$356.93	$238.25	$118.69	$239.00	$17,025.13
3	$356.93	$239.89	$117.05	$356.05	$16,785.24
4	$356.93	$241.54	$115.40	$471.44	$16,543.71
5	$356.93	$243.20	$113.74	$585.18	$16,300.51
6	$356.93	$244.87	$112.07	$697.25	$16,055.64
7	$356.93	$246.55	$110.38	$807.63	$15,809.09
8	$356.93	$248.25	$108.69	$916.32	$15,560.84
9	$356.93	$249.95	$106.98	$1,023.30	$15,310.89
10	$356.93	$251.67	$105.26	$1,128.56	$15,059.22
11	$356.93	$253.40	$103.53	$1,232.09	$14,805.82
12	$356.93	$255.14	$101.79	$1,333.88	$14,550.67
13	$356.93	$256.90	$100.04	$1,433.92	$14,293.77
14	$356.93	$258.66	$98.27	$1,532.19	$14,035.11
15	$356.93	$260.44	$96.49	$1,628.68	$13,774.66
16	$356.93	$262.23	$94.70	$1,723.38	$13,512.43
17	$356.93	$264.04	$92.90	$1,816.28	$13,248.39
18	$356.93	$265.85	$91.08	$1,907.36	$12,982.54
19	$356.93	$267.68	$89.25	$1,996.62	$12,714.86
20	$356.93	$269.52	$87.41	$2,084.03	$12,445.34
21	$356.93	$271.37	$85.56	$2,169.59	$12,173.97
22	$356.93	$273.24	$83.70	$2,253.29	$11,900.73
23	$356.93	$275.12	$81.82	$2,335.11	$11,625.62
24	$356.93	$277.01	$79.93	$2,415.03	$11,348.61
25	$356.93	$278.91	$78.02	$2,493.06	$11,069.69
26	$356.93	$280.83	$76.10	$2,569.16	$10,788.86

15. What is the maximum amount of federal subsidized and unsubsidized loans per year for an independent student in a graduate program?

16. Braden takes out a private student loan for $6,250 in his second year of his undergraduate IT program. He begins paying the loan after 39 months. If Braden's interest rate was 8%, how much is his balance when he begins to pay off the loan?

17. The billing cycle for Therese's credit card is July 7 to August 6 (31 days). The interest rate on the card is 17.9%. The balance on her card at the start of the billing cycle is $925.00. The table below has her transactions for the billing cycle. What is her balance due at the end of the cycle?

Date	Activity	Amount
7-Jul	Billing Date Balance	$925.00
10-Jul	Child Care	$850.00
10-Jul	Gasoline	$51.00
12-Jul	Food	$135.50
18-Jul	Payment	$750.00
18-Jul	Pizza	$35.00
18-Jul	Gift	$25.00
21-Jul	Food	$121.75
30-Jul	Gasoline	$47.50
1-Aug	Department store	$173.00
4-Aug	Food	$98.00

18. The cost of a car is $45,750. The residual price after 2 years is $28,822.50. What is the monthly depreciation for the car?

19. Find the payment for a 3-year lease on a car if the price of the car is $30,000, the residual price of the car is $20,000, and the APR is 8.5%.

20. What are the monthly payments, excluding escrow, for a 20-year mortgage for $131,500 at 5.15% interest?

21. What are the monthly payments, including escrow, for a 20-year mortgage for $131,500 at 5.15% interest, with property taxes of $3,170 per year and homeowner's insurance of $715 every 6 months?

22. If a person earns $2700 gross on a paycheck, how much is taken out in FICA taxes?

23. Find the taxable income for a person who has $81,200 in earnings, $1,250 reported on a 1099-INT for a savings account, $5,400 reported on a 1099-MISC for a contract job they did, paid $5,250 in mortgage interest, paid $4,700 in property taxes, and has $4,200 in bond interest.

24. Find the federal income tax owed by a person filing single who has a taxable income of $114,750. Use the table below to determine the taxes.

Bracket	Lower Income Limit	Upper Income Limit	Tax Rate
1	0	$10,275	10%
2	$10,276	$41,775	12%

Bracket	Lower Income Limit	Upper Income Limit	Tax Rate
3	$41,776	$89,075	22%
4	$89,076	$170,050	24%
5	$170,051	$215,950	32%
6	$215,951	$539,900	35%
7	$539,901		37%

25. If a person owes $20,000 in federal income tax, but qualifies for $4,000 in child tax credit and $8,000 for child and dependent care credit, how much federal tax does the person actually pay?